Lathi 20题练习册

对应 Boulet 前四章

原书英文题干、公式与必要图示 · 无答案版

记号：D = d/dt；E 为前移算子，E{y[n]} = y[n+1]；j^2 = -1
u 为单位阶跃，δ 为单位冲激；LTIC / LTID 分别指连续 / 离散时间 LTI 系统

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来源：B. P. Lathi / R. A. Green, Linear Systems and Signals, 3rd ed. (2018)
中文主题仅用于课程对应；题目要求以原书英文为准。

信号平均功率与均方根

Lathi 1.1-6 | 原书第 137 页

Boulet 第1章：Finite-Energy and Finite-Power Signals， p.21

1.1-6 Determine the power and the rms value for each of the following signals:

(a) $5+10\cos(100t+\pi/3)$

(b) $10\cos(100t+\pi/3)+16\sin(150t+\pi/5)$

(c) $(10+2\sin 3t)\cos 10t$

(d) $10\cos 5t\cos 10t$

(e) $10\sin 5t\cos 10t$

(f) $e^{j\alpha t}\cos\omega_0 t$

演算区

时间反转、平移与尺度变换

Lathi 1.2-6 | 原书第 139 页

Boulet 第1章：Transformations of the Time Variable, p.4

1.2-6 Consider the signal $x(t) = 2^{-tu(t)}$, where $u(t)$ is the unit step function.

(a) Accurately sketch $x(t)$ over $(-1 \leq t \leq 1)$.

(b) Accurately sketch $y(t) = 0.5x(1 - 2t)$ over $(-1 \leq t \leq 1)$.

演算区

周期、非周期与叠加信号的功率

Lathi 1.3-5 | 原书第 140 页

Boulet 第1章：Periodic Signals, p.8；Finite-Energy and Finite-Power Signals, p.21

1.3-5 Given $x_1(t) = \cos(t)$, $x_2(t) = \sin(\pi t)$, and $x_3(t) = x_1(t) + x_2(t)$.

(a) Determine the fundamental periods T_1 and T_2 of signals $x_1(t)$ and $x_2(t)$.

(b) Show that $x_3(t)$ is not periodic, which requires $T_3 = k_1 T_1 = k_2 T_2$ for some integers k_1 and k_2.

(c) Determine the powers P_{x_1}, P_{x_2}, and P_{x_3} of signals $x_1(t)$, $x_2(t)$, and $x_3(t)$.

演算区

冲激函数的筛选与缩放性质

Lathi 1.4-6 | 原书第 142 页

Boulet 第1章：Generalized Functions, p.26

1.4-6 Evaluate the following integrals:

(a) $\int_{-\infty}^{\infty} \delta(\tau)x(t-\tau)\,d\tau$

(b) $\int_{-\infty}^{\infty} x(\tau)\delta(t-\tau)\,d\tau$

(c) $\int_{-\infty}^{\infty} \delta(t)e^{-j\omega t}\,dt$

(d) $\int_{-\infty}^{\infty} \delta(2t-3)\sin \pi t\,dt$

(e) $\int_{-\infty}^{\infty} \delta(t+3)e^{-t}\,dt$

(f) $\int_{-\infty}^{\infty} (t^3+4)\delta(1-t)\,dt$

(g) $\int_{-\infty}^{\infty} x(2-t)\delta(3-t)\,dt$

(h) $\int_{-\infty}^{\infty} e^{(x-1)}\cos\left[\frac{\pi}{2}(x-5)\right]\delta(x-3)\,dx$

演算区

线性系统与非线性系统的判断

Lathi 1.7-1 | 原书第 144 页

Boulet 第1章：System Models and Basic Properties, p.34

1.7-1 For the systems described by the following equations, with the input $x(t)$ and output $y(t)$, determine which of the systems are linear and which are nonlinear.

(a) $\dfrac{dy(t)}{dt} + 2y(t) = x^2(t)$

(b) $\dfrac{dy(t)}{dt} + 3ty(t) = t^2x(t)$

(c) $3y(t) + 2 = x(t)$

(d) $\dfrac{dy(t)}{dt} + y^2(t) = x(t)$

(e) $\left(\dfrac{dy(t)}{dt}\right)^2 + 2y(t) = x(t)$

(f) $\dfrac{dy(t)}{dt} + (\sin t)y(t) = \dfrac{dx(t)}{dt} + 2x(t)$

(g) $\dfrac{dy(t)}{dt} + 2y(t) = x(t)\dfrac{dx(t)}{dt}$

(h) $y(t) = \displaystyle\int_{-\infty}^{t} x(\tau)\,d\tau$

演算区

卷积的时间尺度变换性质

Lathi 2.4-10 | 原书第 227 页

Boulet 第2章：Continuous-Time LTI Systems: The Convolution Integral，p.67；Properties of Linear Time-Invariant Systems，p.74

2.4-10 If $x(t) * g(t) = c(t)$, then show that $x(at) * g(at) = |1/a|c(at)$. This *time-scaling property* of convolution states that if both $x(t)$ and $g(t)$ are time-scaled by a, their convolution is also time-scaled by a (and multiplied by $|1/a|$).

演算区

连续时间图解卷积与分段积分

Lathi 2.4-12 | 原书第 227 页

Boulet 第2章：Continuous-Time LTI Systems: The Convolution Integral, p.67

2.4-12 Suppose an LTIC system has impulse response $h(t) = (1-t)[u(t) - u(t-1)]$ and input $x(t) = u(-t-1) + u(t-1)$. Use the graphical convolution procedure to determine $y_{zsr}(t) = x(t) * h(t)$. Accurately sketch $y_{zsr}(t)$. When solving for $y_{zsr}(t)$, flip and shift $h(t)$, explicitly show all integration steps, and simplify your answer.

演算区

因果指数信号的卷积

Lathi 2.4-13 | 原书第 227 页

Boulet 第2章：Continuous-Time LTI Systems: The Convolution Integral, p.67

2.4-13 Using direct integration, find $e^{-at}u(t) * e^{-bt}u(t)$.

演算区

离散时间卷积和与时间平移

Lathi 3.8-7 | 原书第 322 页

Boulet 第2章：Discrete-Time LTI Systems: The Convolution Sum， p.54

3.8-7 Find the (zero-state) response $y[n]$ of an LTID system if the input $x[n] = (3)^{n+2}u[n+1]$, and

$$h[n] = [(2)^{n-2} + 3(-5)^{n+2}]u[n-1]$$

演算区

离散LTI系统的因果性、稳定性与零状态响应

Lathi 3.9-5 | 原书第 327 页

Boulet 第2章：Discrete-Time LTI Systems: The Convolution Sum，p.54；Properties of Linear Time-Invariant Systems，p.74

3.9-5 A discrete-time LTI system has impulse response given by

$$h[n] = \delta[n] + \left(\tfrac{1}{3}\right)^n u[n-1]$$

(a) Is the system stable? Is the system causal? Justify your answers.
(b) Plot the signal $x[n] = u[n-3] - u[n+3]$.
(c) Determine the system's zero-state response $y[n]$ to the input $x[n] = u[n-3] - u[n+3]$. Plot $y[n]$ over $(-10 \le n \le 10)$.

演算区

连续系统的特征方程、模态与冲激响应

Lathi 2.3-1 | 原书第 224、225 页

Boulet 第3章：Impulse Response of a Differential LTI System， p.101；Characteristic Polynomials and Stability， p.112

2.3-1 Determine the characteristic equation, characteristic modes, and impulse response $h(t)$ for each of the following real LTIC systems. Since the systems are real, express each $h(t)$ using only real terms (i.e., no j's should appear in your answers).

(a) $(D^2+1)\{y(t)\} = 2D\{x(t)\}$

(b) $(D^3+D)\{y(t)\} = (2D^3+1)\{x(t)\}$

(c) $\frac{d^2}{dt^2}y(t) + 2\frac{d}{dt}y(t) + 5y(t) = 8x(t)$

演算区

含输入导数的一阶系统冲激响应

Lathi 2.3-4 | 原书第 225 页

Boulet 第3章：Impulse Response of a Differential LTI System，p.101

原题引用2.3-2的作答要求：求方程所描述系统的单位冲激响应 h(t)。

2.3-4 Repeat Prob. 2.3-2 for the first-order allpass filter specified by the equation

$$(D+1)y(t) = -(D-1)x(t)$$

演算区

差分方程的递推解：冲激、阶跃与零输入响应

Lathi 3.5-1 | 原书第 319 页

Boulet 第3章：Causal LTI Systems Described by Difference Equations, p.96

3.5-1 An LTID system is described by a constant coefficient linear difference equation $2y[n] + 2y[n-1] = x[n-1]$.

(a) Express this system in standard advance operator form.

(b) Using recursion, determine the first 5 values of the system impulse response $h[n]$.

(c) Using recursion, determine the first 5 values of the system zero-state response to input $x[n] = 2u[n]$.

(d) Using recursion, determine for $(0 \leq n \leq 4)$ the system zero-input response if $y[-1] = 1$.

演算区

离散系统的阶数、特征模态与冲激响应

Lathi 3.7-3 | 原书第 321 页

Boulet 第3章：Impulse Response of a Difference LTI System， p.109

3.7-3 Consider a DT system with input $x[n]$ and output $y[n]$ described by the difference equation

$$4y[n+1]+y[n-1]=8x[n+1]+8x[n]$$

(a) What is the order of this system?
(b) Determine the characteristic mode(s) of the system.
(c) Determine a closed-form expression for the system's impulse response $h[n]$.

演算区

特征多项式与系统稳定性

Lathi 2.5-1 | 原书第 233 页

Boulet 第3章：Characteristic Polynomials and Stability of Differential and Difference Systems, p.112

2.5-1 Explain, with reasons, whether the LTIC systems described by the following equations are (i) stable or unstable in the BIBO sense; (ii) asymptotically stable, unstable, or marginally stable. Assume that the systems are controllable and observable.

(a) $(D^2+8D+12)y(t)=(D-1)x(t)$

(b) $D(D^2+3D+2)y(t)=(D+5)x(t)$

(c) $D^2(D^2+2)y(t)=x(t)$

(d) $(D+1)(D^2-6D+5)y(t)=(3D+1)x(t)$

演算区

谐波、基频与三角傅里叶级数系数

Lathi 6.1-5 | 原书第 669 页

Boulet 第4章：Linear Combinations of Harmonically Related Complex Exponentials，p.132；Determination of the Fourier Series Representation，p.134

6.1-5 Determine the trigonometric Fourier series coefficients a_n and b_n for the following signals. In each case, also determine the signals' fundamental radian frequency ω_0. No integration is required to solve this problem.

(a) $x_a(t) = \cos(3\pi t)$

(b) $x_b(t) = \sin(7\pi t)$

(c) $x_c(t) = 2 + 4\cos(3\pi t) - 2j\sin(7\pi t)$

(d) $x_d(t) = (1+j)\sin(3\pi t) + (2-j)\cos(7\pi t)$

(e) $x_e(t) = \sin(3\pi t + 1) + 2\cos(7\pi t - 2)$

(f) $x_f(t) = \sin(6\pi t) + 2\cos(14\pi t)$

演算区

分段周期信号的复指数傅里叶级数

Lathi 6.3-2 | 原书第 672 页

Boulet 第4章：Determination of the Fourier Series Representation of a Continuous-Time Periodic Signal, p.134

6.3-2 A 2π-periodic signal $x(t)$ is specified over one period as

$$x(t) = \begin{cases} \frac{1}{A}t & 0 \le t < A \\ 1 & A \le t < \pi \\ 0 & \pi \le t < 2\pi \end{cases}$$

Sketch $x(t)$ over two periods from $t = 0$ to 4π. Show that the exponential Fourier series coefficients D_n for this series are given by

$$D_n = \begin{cases} \frac{2\pi - A}{4\pi} & n = 0 \\ \frac{1}{2\pi n}\left(\frac{e^{-jAn} - 1}{An} + je^{-jn\pi}\right) & n \neq 0 \end{cases}$$

演算区

傅里叶级数的奇偶与共轭对称性质

Lathi 6.3-8 | 原书第 673 页

Boulet 第4章：Properties of Continuous-Time Fourier Series, p.139

6.3-8 Let periodic signal $x(t)$ have exponential Fourier series spectrum D_n. Prove the following properties.

(a) If $x(t)$ has even symmetry, then D_n also has even symmetry.

(b) If $x(t)$ has odd symmetry, then D_n also has odd symmetry.

(c) If $x(t)$ is real, then D_n is conjugate symmetric ($D_n = D^*_{-n}$).

(d) If $x(t)$ is imaginary, then D_n is conjugate antisymmetric ($D_n = -D^*_{-n}$).

演算区

周期输入经过LTI滤波系统后的输出

Lathi 6.4-3 | 原书第 675 页

Boulet 第4章：Steady-State Response of an LTI System to a Periodic Signal, p.155

6.4-3 Consider a $T_0 = 1$ periodic signal $x(t)$ defined as

$$x(t) = \begin{cases} 1 - t^2 & 0 < t \leq 1 \\ x(t+1) & \forall t \end{cases}$$

(a) Sketch $x(t)$ for $-2 \leq t \leq 2$.
(b) Determine D_n, the exponential Fourier series spectrum of $x(t)$.
(c) If $x(t)$ is applied to an ideal bandpass filter with a 1 Hz passband centered at 3 Hz, determine the output $y(t)$.

演算区

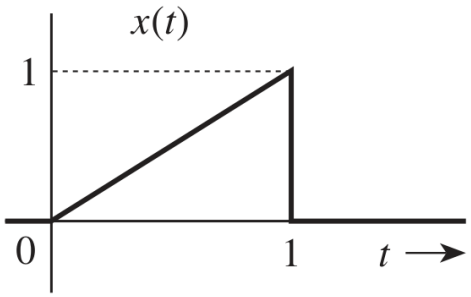
傅里叶级数的有限项近似与误差能量

Lathi 6.5-6 | 原书第 677 页

Boulet 第4章：Optimality and Convergence of the Fourier Series，p.144；Parseval Theorem，p.150

所需原图 P6.5-4(a) 已附在题干下方；本题只使用图(a)，不使用图(b)。

6.5-6 Represent the signal $x(t)$ shown in Fig. P6.5-4a over the interval from 0 to 1 by a trigonometric Fourier series of fundamental frequency $\omega_0 = 2\pi$. Compute the error energy in the representation of $x(t)$ by only the first N terms of this series for $N = 1, 2, 3$, and 4.



(a)

Figure P6.5-4

演算区